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Machine learning-based optimization of alginate, guar gum, and pectin-based edible coatings for extended strawberry shelf life

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ABSTRACT

Strawberries deteriorate rapidly under warm, humid conditions, and to make things worse, in many tropical settings refrigerated logistics are limited. Edible biopolymer coatings offer a low-cost method to slow quality loss, but multicomponent formulations behave nonlinearly. In this study, ten types of coatings were developed and evaluated for their effects on the physicochemical (weight loss, firmness, total soluble solids (TSS), pH, color change (ΔE)) and biochemical properties (total phenolic content (TPC), antioxidant property (DPPH) of strawberries over a 6-day storage period at 25 °C (298.15K). The optimal formulation, consisting of 0.56 % (w/v) alginate, 0.33 % (w/v) guar gum, and 2.11 % (w/v) pectin, shows 7 % weight loss, 586.34 N firmness, a pH of 4.08, and a ΔE of 4.32 according to our machine learning models. Additionally, two formulation families emerged from the optimization: pectin-rich coatings extended shelf-life with lower weight loss and restrained color change, while alginate-rich coatings maintained better firmness. All machine learning models (Random Forest (RF), Support Vector Regression (SVR), and eXtreme Gradient Boosting (XGB)) achieved high predictive accuracy ($R^2 > 0.97$).

1. Introduction

Strawberries (*Fragaria x ananassa*) are widely consumed and cultivated fruits across the globe, cherished for their sweet, juicy flavor and numerous health benefits (Moghadas et al., 2025). Originating from the hybridization of North American species, strawberry cultivation has now spread to temperate and subtropical regions, with global production surpassing 9.6 million tons in 2022 (Carole et al., 2024). These fruits are rich in micronutrients, particularly folate, vitamin C, vitamin E, phenolic compounds, flavonoids, and essential minerals like calcium, iron, and phosphorus, which contribute to a healthy heart and give anti-mutagenic properties (Basu et al., 2014). However, strawberries are vulnerable to storage conditions such as temperature, atmospheric composition (e.g., CO2 levels), which strongly influence strawberries' firmness, TSS, pH, and titratable acidity (TA) (Chandra et al., 2015). Biochemical processes during maturation and senescence involve enzymatic activities (like polygalacturonase and pectin methyl esterase) that degrade cell walls, leading to texture softening and changes in fiber content. These processes also contribute to changes in anthocyanin pigments, organic acids, and sugar content that impact flavor and appearance. Microbial activity can also increase pH by metabolizing organic compounds, contributing to deterioration.

While the use of improved cultivars and refrigeration has enhanced strawberry quality in East and Southeast Asia, these warmer areas are still encountering increased challenges due to rapid postharvest deterioration (Nishizawa, 2021). Many warmer regions still face challenges due to limited and expensive cold-chain infrastructure for post-harvest transportation, leading to higher greenhouse gas emissions and increased energy consumption. (Wang et al., 2025; Yu et al., 2025). As the global demand for sustainable solutions intensifies, the food industry is exploring innovative alternatives, such as biopolymer-based edible coatings made from alginate, chitosan, and carrageenan etc. (Kumar et al., 2021). These coatings are not only cost-effective but also environmentally sustainable, addressing the critical need for green technologies that minimize waste and resource consumption (N. Kumar et al., 2023; Muley & Singhal, 2020).

Edible coatings have emerged as a promising solution to this problem. They act as semi-permeable barriers to gases and moisture, and contain added functionalities, thereby slowing down ripening and

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Abbreviations:

ANOVA	Analysis of Variance
DPPH	2,2-Diphenyl-1-picrylhydrazyl
DE	Differential Evolution
PSO	Particle Swarm Optimization
DF	Degree of Freedom
GAE	Gallic Acid Equivalents
MAE	Mean Absolute Error
MSE	Mean Squared Error
RF	Random Forest
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
SVR	Support Vector Regression
TSS	Total Soluble Solids
TPC	Total Phenolic Content
XGB	eXtreme Gradient Boosting
a*	Redness – Greenness
b*	Blueness – Yellowness
L*	Lightness
ΔE	Total Color Difference

microbial spoilage (Falcao-Rodrigues et al., 2007; Kumar et al., 2022; Kumarihami et al., 2023). Studies have shown that these coatings not only enhance shelf life but also reduce the reliance on refrigeration (N. Kumar & Neeraj, 2019; Prasad et al., 2025), offering a sustainable solution in regions with limited cold-chain infrastructure, reinforcing their potential to reduce both food waste and environmental impact (Trajkovska Petkoska et al., 2021a).

A linear polysaccharide derived from brown seaweed, sodium alginate, is frequently used in food coating due to its gelling properties, which provide a strong barrier to moisture and gas transmission, essential for slowing down the ripening process (Makhathini et al., 2025). Alginate forms firm gels in the presence of calcium ions, creating an effective barrier to moisture and gas transmission (Nair et al., 2020). However, alginate alone tends to be brittle and inflexible, which limits its barrier properties and ability to control metabolic activities (Rastegar et al., 2019). To address this, this study mixes alginate with guar gum and pectin. Guar gum, derived from legume seeds, offers excellent water-holding capacity, emulsifying properties, and flexibility (Ruelas-Chacon et al., 2017). They also improve texture, reduce weight loss, and retard staling in bakery products, making guar gum an effective additive for preserving freshness and extending shelf life in food coatings (Mudgil et al., 2014). Similarly, pectin, a complex polysaccharide found in plant cell walls, has gelling, stabilizing, and antioxidant properties, making it an excellent co-component (S. Kumar et al., 2023). It contributes antioxidant properties and film-forming ability, which can protect food products from oxidative damage and enhance mechanical strength in edible coatings. Pectin helps maintain product quality by controlling moisture migration and providing structural flexibility. Its gelation properties reinforce the film matrix, thereby improving barrier functions against moisture and gas transmission (Yazıcıoğlu & Siyasal, 2024).

Formulating such multicomponent coatings presents significant challenges, as the complex interactions between ingredients often lead to nonlinear and unpredictable effects on fruit quality (Trajkovska Petkoska et al., 2021b). Relying solely on trial-and-error methods is inefficient and may overlook optimal compositions. To overcome these challenges, advanced computational approaches such as machine learning, deep learning, and computer vision are being employed (da Silva Rios et al., 2022; Rahman et al., 2024; Sha et al., 2025). ML algorithms are capable of modeling these complex and nonlinear relationships. These models learn patterns from historical data and can predict outcomes under varying conditions (Chen et al., 2022). For example, ML models can be trained to predict what physicochemical properties will be under certain situations, and optimize what processing would maximize the quality in that scenario (Wang et al., 2024). In recent times, ML-based optimization is used to enable more precise control over the coating composition and storage conditions, to address crucial risks under resource-limited settings (Menichetti et al., 2023).

In this study, a machine learning models for multi-objective optimization were developed, trained on augmented data to predict the ideal coating formulation for strawberries. The formulation includes three additives (alginate, guar gum, and pectin) to optimize key properties such as firmness, TSS, moisture retention, and bioactive properties like TPC and DPPH. These properties are critical for designing effective edible coatings. By incorporating augmented data into the machine learning training process, the study improves model performance and demonstrates how data augmentation can capture complex relationships within real-world datasets.

2. Materials and methods

2.1. Materials & chemicals

Strawberries (Fragaria x ananassa, var. Robi 3) were collected from Akafuji Agrotechnologies, Boalia, Rajshahi, Bangladesh, with no visible deterioration or damage. Strawberries were refrigerated aboard a temperature-controlled truck at 0–2 °C (273.15K–275.15K), with 90–95 % relative humidity, and packed in ventilated punnets to minimize moisture loss and preserve quality. Alginate, guar gum, pectin, and food-grade gelatin were obtained from Merck (Darmstadt, Germany). Folin-Ciocalteau reagent, sodium carbonate, gallic acid, and 2,2-diphenyl-1-picrylhydrazyl (DPPH) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Methanol and acetone were bought from Merck (Darmstadt, Germany). Single-distilled water was used for all purposes needed.

2.2. Simplex-centroid mixture design

An augmented simplex-centroid design was used by expanding the standard 7-point design by adding 3 intermediate blend points to cover the experimental space better and capture nonlinear interactions (Supplementary Fig. 1) (de Souza et al., 2021). Ten treatments were performed with varying proportion of alginate (x_1), guar gum (x_2), and pectin (x_3) as follows: 0 (0 %), 1/6 (16.67 %), 1/3 (33 %), 1/2 (50 %), 2/3 (66.67 %), and 1 (100 %), as shown in Table 1.

The total solids concentration of 3 % (w/v, 30 g L^{-1}) was selected based on preliminary trials, where higher concentrations (>3 %) resulted in excessively viscous solutions, making uniform application difficult, while lower concentrations (≈2 %) led to coatings prone to surface runoff and poor adhesion. This concentration aligns with literature precedents for similar polysaccharide-based coatings (de Souza et al., 2021).

Table 1
Matrix of the simplex-centroid mixture design and the biopolymer combinations (coded and real values).

Runs	Treatment		
	Alginate (x1)	Guar Gum (x2)	Pectin (x3)
1	1 (3.0 %)	0 (0.0 %)	0 (0.0 %)
2	0 (0.0 %)	1 (3.0 %)	0 (0.0 %)
3	0 (0.0 %)	0 (0.0 %)	1 (3.0 %)
4	1/2 (1.5 %)	1/2 (1.5 %)	0 (0.0 %)
5	1/2 (1.5 %)	0 (0.0 %)	1/2 (1.5 %)
6	0 (0.0 %)	1/2 (1.5 %)	1/2 (1.5 %)
7	2/3 (2.0 %)	1/6 (0.5 %)	1/6 (0.5 %)
8	1/6 (0.5 %)	2/3 (2.0 %)	1/6 (0.5 %)
9	1/6 (0.5 %)	1/6 (0.5 %)	2/3 (2.0 %)
10	1/3 (1.0 %)	1/3 (1.0 %)	1/3 (1.0 %)

2.3. Preparation of coating solutions

Coating solutions were prepared according to (Gull et al., 2024) with some modifications. For each concentration defined by the simplex centroid method, polymers were dissolved in distilled water at 70 °C (343.15K) using a magnetic stirrer (MS7-H550-S DLAB, USA) at 150 rpm (no vortex) until they were completely dissolved. Food-grade gelatin (2 % w/v) was added to each solution to enhance coating adherence. Samples were sonicated in an ultrasonic bath (Bransonic 2510-DTH, 40 kHz; 2.84 L tank; Emerson/Branson, USA) for 30 min (1800s) with the bath maintained at 40 °C (313.15K) to eliminate air bubbles and allowed to rest at room temperature for an additional 60 min (3600s) before applying. Fig. 1 illustrates the flow diagram of the preparation and application process of edible coatings.

2.4. Application of edible coating

Strawberries were rinsed with distilled water, air-dried, and then immersed in the prepared coating solutions for 3 min (180s). Excessive coating was drained through sanitized stainless-steel sieves. The fruits were then placed in a sterile polypropylene tray and stored at 25 °C (298.15K) with 85 % relative humidity for up to 6 days. Control samples underwent the same procedure but were dipped only in a 2 % gelatin solution. A schematic diagram of the strawberry coating process is shown in Fig. 2.

2.5. Sample extraction

For analysis, the strawberry samples were collected at 0, 2, 4, and 6 days and lightly rinsed to remove residual coating before extraction. Then they were homogenized using a mixer grinder (MX-AC400, Panasonic, Japan) and filtered to obtain a puree (Wani et al., 2021). After that, 10 mL of previously made 80 % (v/v) acetone was combined with 1 g of the ground material, and the mixture was incubated for 90 min (5400s) at 20 °C (293.15K). The crude extract was then centrifuged for 15 min (900s) at 4000 rpm (5804R, Eppendorf, Germany). The resulting supernatant was collected and kept at −20 °C (253.15K) until further examination.

2.6. Physicochemical properties

2.6.1. Weight loss and firmness

Pre-weighed containers holding the sample were weighed using a semi-analytical balance (ATY-224R, Shimadzu, Japan). Weight loss (%) was calculated as in Eq (1).

$$\text{Weight Loss}\ (\%) = ((\textit{Initial Weight} - \textit{Final Weight})\ /\ (\ \textit{Initial weight})\ ) \times 100 \quad (1)$$

Firmness was measured using a texture analyzer (AGROSTA Fruit, Agrosta, France) equipped with a 60° cone (Rico et al., 2007) and expressed in N.

2.6.2. Total soluble solids (TSS) and pH

A hand refractometer (0–35° Brix, Atago, Japan) was then used to measure the total soluble solids (TSS). A digital pH meter (HI-2211, Hanna Instrument, USA) was used to measure the pH. In between tests, the electrode was cleaned and calibrated using a buffer solution.

2.6.3. Color

Color measurements (L*, a*, and b*) were taken using a reflectance colorimeter (PCE-CSM4, PCE Instruments, Germany) calibrated against a white surface. Readings were taken directly from strawberry clusters at randomly selected spots. The total color difference (ΔE), which quantifies the overall perceptible change in color, was then calculated as in Eq (2).

$$\Delta E = \left[\left(L_{sample} - L_{standard}\right)^2 + \left(a_{sample} - a_{standard}\right)^2 + \left(b_{sample} - b_{standard}\right)^2 \right]^{1/2} \quad (2)$$

2.7. Bioactive properties

2.7.1. Total phenolic content (TPC)

Total phenolic content was measured using the Folin-Ciocalteu method with slight modification (Singleton et al., 1999). Each sample contained 20 μL of the extract, 1.58 mL of distilled water, and 100 μL of Folin-Ciocalteu reagent. The mixture was then allowed to react for 8 min. The mixture was immediately vortexed after 300 μL of sodium carbonate solution was added. Then, for half an hour, the mixture was incubated at 40 °C (313.15k) in the dark. Using a spectrophotometer (UV1800, Shimadzu, Japan), the absorbance was measured at 765 nm following incubation. A blank was made using the same adjustment procedure, but instead of using the sample, distilled water was used. A standard calibration curve was then produced using gallic acid solutions at various concentrations. The TPC results were expressed as gallic acid equivalents, μg GAE/g FW.

2.7.2. Antioxidant activity

The antioxidant activity was measured using the DPPH radical scavenging method, following Rasera et al. (2019) with slight modifications. Absorbance was measured at 517 nm using a UV spectrophotometer (UV1800, Shimadzu, Japan). 1.4 mL of DPPH (0.1 mM) solution was combined with 100 μL of strawberry extract, and the mixture was left in the dark for 30 min before measuring the absorbance. For the blank, water was used in place of the extract. DPPH inhibition (%) was then calculated as in Eq (3).

$$\text{DPPH Inhibition}\ (\%) = \left(\left(A_{control} - A_{sample}\right)/\left(A_{control}\right)\right) \times 100 \quad (3)$$

$A_{control}$ = Absorbance of DPPH solution without the sample.

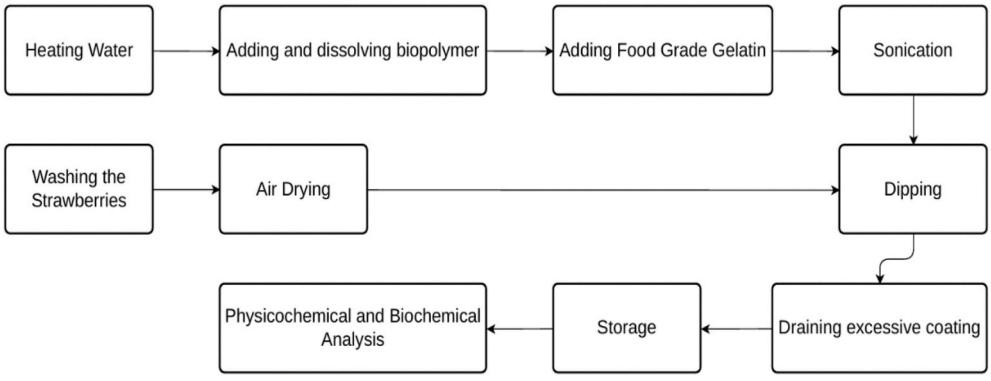


Fig. 1. Flow diagram illustrating the preparation and application of edible coating solutions.

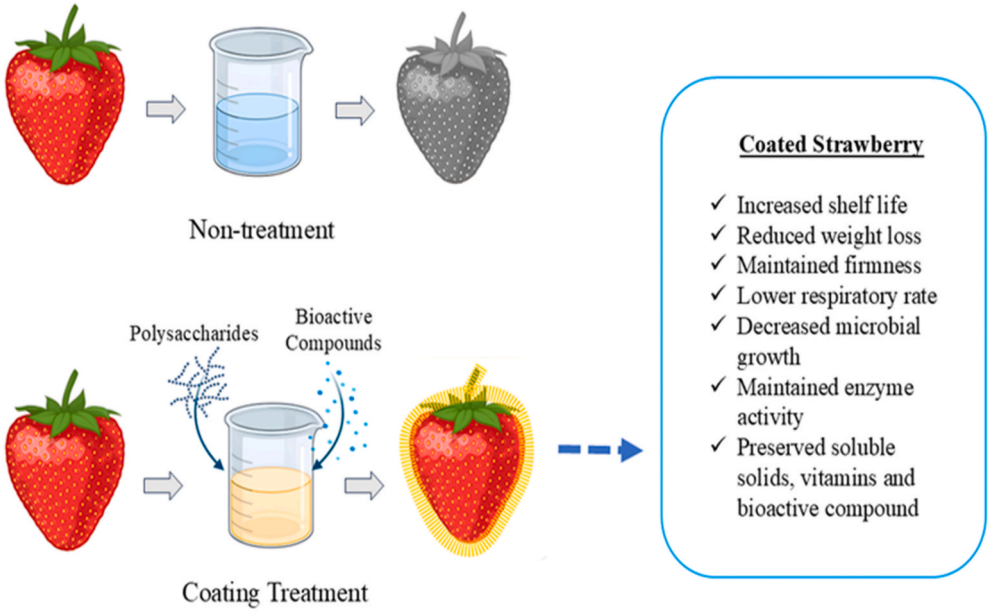


Fig. 2. Schematic diagram of edible coating on strawberry.

A_{sample} = Absorbance of DPPH solution with sample.

2.8. Data augmentation

To enhance the robustness and generalizability of the predictive models, the experimental dataset was expanded through data augmentation. This approach is commonly used in systems where data availability is limited. For data augmentation, a process described by Abookleesh et al. (2025) was followed. A process diagram of the data augmentation and machine learning approach is presented in Fig. 3.

The original dataset, consisting of triplicate measurements, was averaged based on the storage days and coating types, and standard deviations were calculated to capture variability. Controlled random noise was then added to the averaged values, with the noise sampled from a normal distribution where the standard deviation was set to one-third of the observed variability of the dataset. This approach generated 99 synthetic samples, which were combined with the original data to form an augmented dataset.

2.9. Numerical modeling

2.9.1. Model training and tuning

Three supervised learning algorithms, RF, SVR, and XGB, were trained to predict the physicochemical attributes based on predictor variables: storage time and the concentrations of alginate, guar gum, and pectin.

To develop a robust and generalizable predictive model, an 80/20 train-test split was performed on the augmented dataset. Hyperparameter optimization was then carried out using GridSearchCV with RepeatedKFold (5 folds × 10 repeats, random state = 4) on the training data. Performance metrics such as R^2, MAE, and RMSE were computed across folds. The optimal hyperparameter set for each model was recorded for further use.

Among the models employed, SVR was selected for its ability to handle small datasets with nonlinear relationships. The optimal hyperparameters found: C = 1000, epsilon = 0.01, gamma = 0.1, and an RBF kernel. RF constructs multiple decision trees from bootstrapped samples to reduce variance, with optimal hyperparameters found: n_estimators = 100, min_samples_split = 2, and min_samples_leaf = 1. XGB uses gradient boosting to sequentially correct residuals, with the best configuration found: learning_rate = 0.1, n_estimators = 1200, max_depth = 7, subsample = 0.5, and gamma = 0.01.

2.9.2. Learning curve analysis

To assess data efficiency and potential overfitting, a learning curve was generated using scikit-learn's 'learning_curve'. Training size was set to 100 % of the training set. At each size, a 5-fold cross-validation was performed, and mean ± SD RMSE were recorded. Errors were reported on the original target scales and averaged uniformly across the multiple outputs; hence, the multi-output RMSE reflects variables with larger numeric scales.

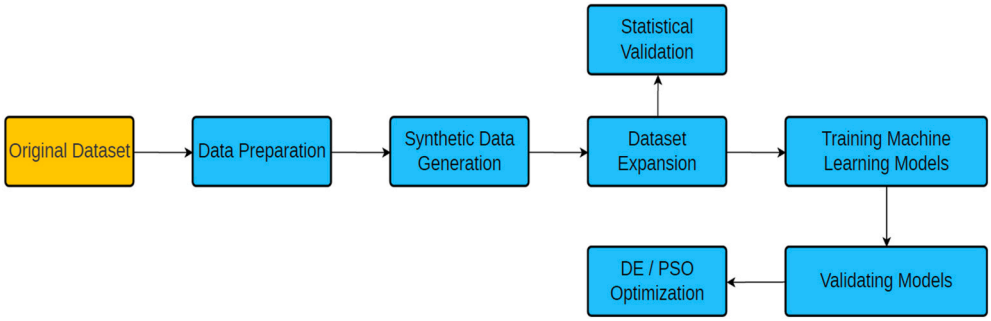


Fig. 3. Data augmentation and machine learning modeling pipeline flow diagram.

2.10. Numerical optimization

The storage duration and coating materials were optimized using the Differential Evolution (DE) and Particle Swarm Optimization (PSO) algorithms. DE is widely recognized as one of the most reliable and flexible population-based search algorithms due to its capacity to handle multi-modal problems and perform global optimization of nonlinear, non-differentiable continuous functions. The algorithm starts with an initial population of potential solutions, similar to other common direct search methods like genetic algorithms and evolutionary tactics (Cristina, 2021). The best formulated solutions that generate a lower objective function are kept in the population by introducing mutations. Particle swarm optimization (PSO), on the other hand, is an optimization method inspired by nature that differs from conventional approaches in that it does not depend on gradients or derivatives of the objective function. Instead, it operates solely using the objective function itself (Cristina, 2021).

To find the best coating formulation, a four-dimensional decision vector x = (d, a, g, p) was defined, and the levels of alginate (a), guar gum (g), and pectin (p) were constrained to be: a + g + p = 3. A penalty term was then introduced to enforce the total composition allowed (Eq. (4))

$$coating\ penalty = 10^3 * |(alginate + guar\ gum + pectin) - 3| \quad 4$$

The goal of the optimization is to extend the storage time of the strawberries while maintaining or improving their quality. To scalarize this multi-objective optimization, a quality score was formulated (Eq. (5)) that combines both desirable and undesirable attributes.

$$quality\ score = 2*firmness + 2*TSS + 1.2|pH - 4.12| - 2*weightloss - 1*\Delta E \quad 5$$

To account for both the coating formulation and the storage time, the total objective function combines the quality score with the coating penalty and a penalty for shorter storage durations. The total objective is given by Eq. (6), which follows:

$$Total\ Objective = -(quality\ score - coating\ penalty + 2*days) \quad 6$$

2.11. Statistical analysis

All experiments were conducted in triplicate to ensure reliability. Statistical analyses and machine learning model training were performed using Python 3.11. A two-way ANOVA was conducted to evaluate the effects of storage duration, coating treatment, and their interaction on the physicochemical and bioactive characteristics of strawberries. Significant differences among means were further analyzed using Duncan's Multiple Range Test (DMRT) at the 5 % significance level. Pearson correlation coefficients were also calculated

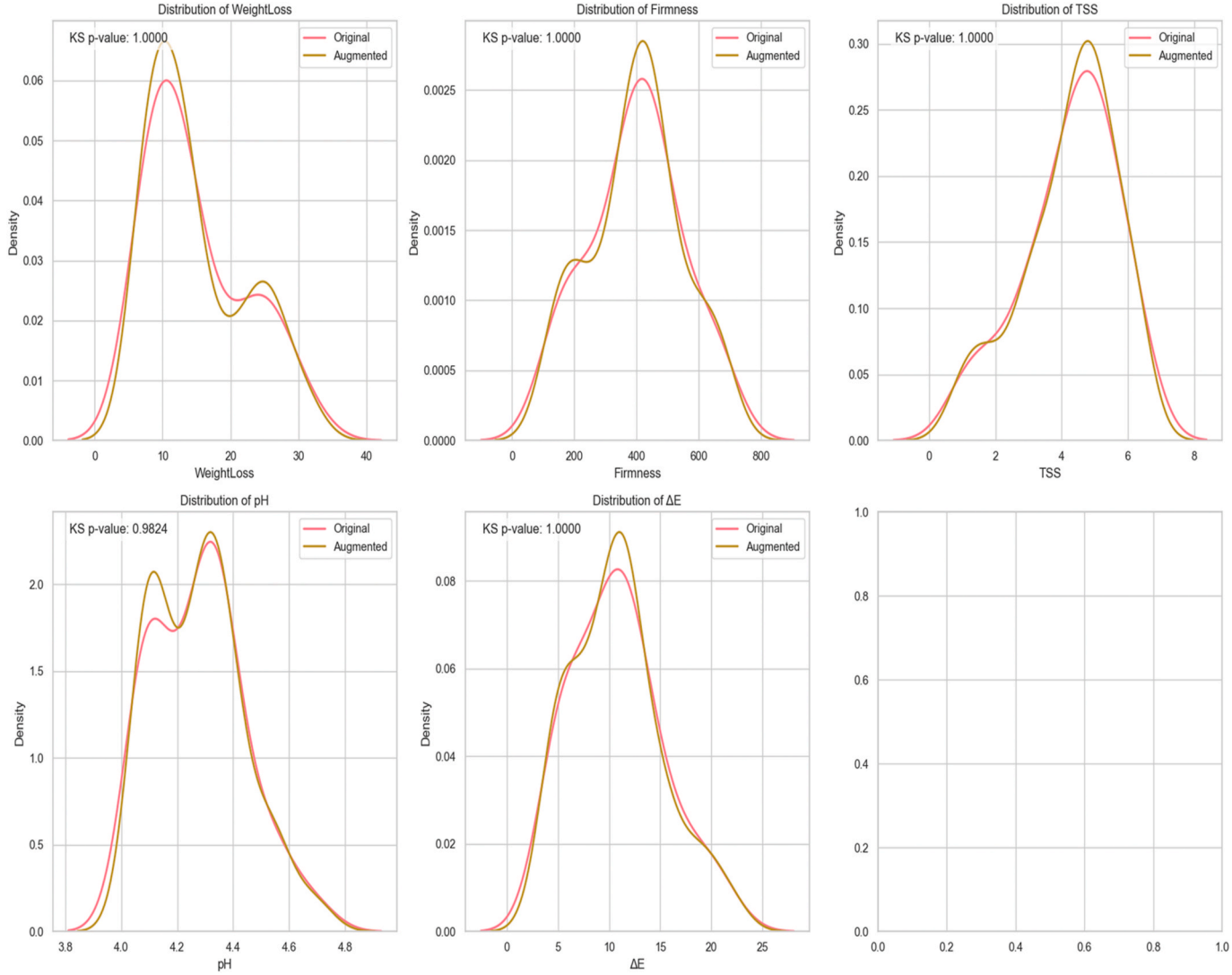


Fig. 4. Distribution comparison of key physicochemical properties of strawberries: Original vs. Augmented data.

among the measured physicochemical and bioactive variables using the corr() method in Python's pandas library and visualized using Seaborn.

3. Results and discussion

3.1. Quality of generated data

Kernel-density overlays of the original and augmented sets show identical overlap for all variables with no visible shifts in central tendency or spread (Fig. 4). Formal two-sample tests were nonsignificant for every variable ($\alpha = 0.05$), and effect sizes were negligible (Cohen's d $\sim$ 0), indicating that the augmentation introduced realistic noise without altering the underlying distributions (see Supplementary Table 1) (Qiu et al., 2025).

Moreover, Pearson correlation patterns were essentially unchanged after augmentation (Fig. 5), showing that negative associations and positive associations were maintained with only trivial differences in magnitude. Preserving these relationships is a key criterion for synthetic/augmented tabular data fidelity because downstream models depend on the covariance structure, not just marginal distributions (Du & Li, 2024).

Taken together, the overlapping KDEs, nonsignificant KS results, negligible effect sizes, and near-identical correlation maps indicate that augmentation retained the statistical properties of the original dataset while adding plausible variability, appropriate for training and validating the predictive models used here.

3.2. Physicochemical properties of fresh strawberries

Table 2 summarizes the initial characteristics of fresh strawberries. The average total soluble solids content was 7.9 °Brix, and a pH of 4.12. The color parameters L*, a*, and b* were recorded as 21.67, 9.51, and 16.23, respectively. The total phenolic content was 718.1 μg GAE/g FW. Edible coatings were applied to slow down the metabolic rate of the strawberries, help retain their nutritional quality, and prolong their shelf life.

3.3. Evaluation of coated strawberry quality

3.3.1. Weight loss

Fig. 6A and B illustrates the percentage weight loss in the samples throughout the 6-day storage period. The experienced weight loss is mainly attributed to surface water evaporation and the breakdown of stored carbohydrates through respiration (Kader, 2002). According to the two-way ANOVA results (Supplementary Table 2), storage duration, coating types, and their interaction significantly influenced weight loss, aligning with the observations made by Rabasco-Vílchez et al. (2024) in their study on grapes.

The control showed the highest weight loss, increasing from 14.72 % on day 2–31.38 % on day 6 ($p < 0.05$, Supplementary Table 5). This considerable moisture loss is due to the thin epidermis and lack of any protective layers to minimize evaporation. The elevated storage temperature (25 °C, 298.15K) also likely accelerated this process (Mdenye et al., 2016). In contrast, treated samples exhibited significantly less weight loss. Treatments 7 and 4 consistently showed low weight loss (12.59 % and 16.38 %, respectively) throughout storage, with Duncan's post-hoc test indicating that these treatments were performing significantly better. The composition of polymers in these coatings likely formed denser polymer networks, enhancing moisture retention and reducing metabolic activity by restricting oxygen exchange through an improved microenvironment (Khodaei et al., 2021).

Treatments 5, 6, and 8 were comparatively less effective but still performed significantly better than the control. According to Duncan's post-hoc test, these treatments showed significant reductions in weight loss compared to the control, but their performance was not as pronounced as treatments 7 and 4 ($p < 0.05$). According to Priyadarshi et al. (2024), this may be due to thin coatings that do not form continuous barriers, leading to only slight reductions in water vapor transmission and respiration. Natural waxes or surface traits of strawberries also likely played a role in influencing coating effectiveness.

3.3.2. Firmness

Firmness is a critical quality attribute, as firmer strawberries are less prone to decay, better withstand handling and transport, and maintain higher sensory appeal (Ren et al., 2023). Supplementary Table 2 shows that storage period, coating types, and their interaction significantly affected firmness value. Both the coated and uncoated strawberries softened over time (Fig. 6C and D). This gradual softening is due to the natural postharvest physiological processes, including the enzymatic breakdown of structural polysaccharides and proteins within the cell wall, leading to tissue weakening (Balic et al., 2014).

In our study, treatment 7 demonstrated the best firmness retention

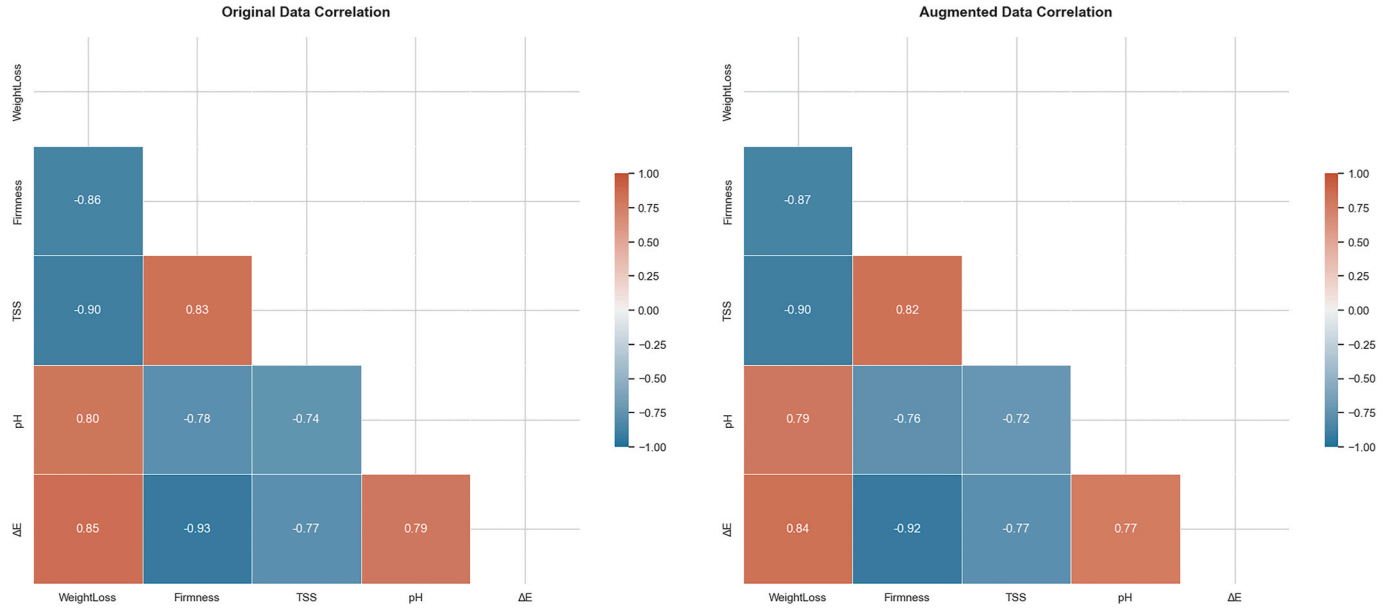


Fig. 5. Pearson correlation of key physicochemical properties of strawberries: Original vs. Augmented data.

Table 2
The physicochemical, color, and bioactive properties of fresh strawberries.

Parameters for fresh Strawberries	Physicochemical Profile			Color			Bioactivity
	Firmness (N/mm)	TSS (°Brix)	pH	l*	a*	b*	TPC (μg GAE/g)
	730.33 ± 1.64	7.9 ± 0.43	4.12 ± 0.02	21.67 ± 1.11	9.51 ± 0.19	16.23 ± 1.02	718.1 ± 2.04

Data are mean ± standard deviation (SD); pH; L = lightness; a* = redness–greenness; b* = blueness–yellowness; TPC = Total Phenolic Content.

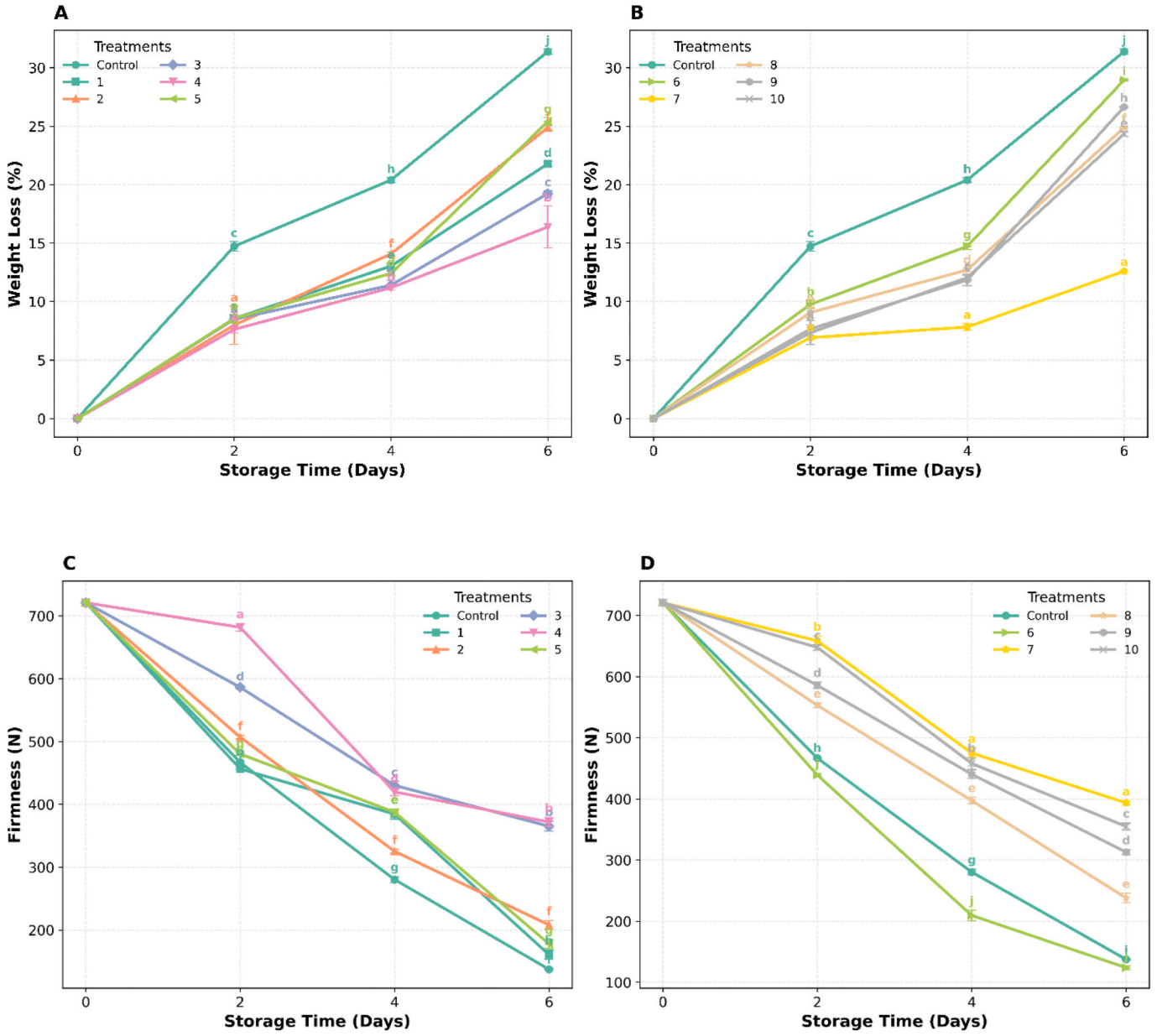


Fig. 6. Effect of different treatments on the weight loss percentage (A–B) and firmness (C–D) of strawberries during 6 days of storage. Data are presented as mean ± standard deviation.

throughout the storage, maintaining higher firmness than other treatments ($p < 0.05$, Supplementary Table 5). Treatments 4 and 10 also maintained high firmness values, although minor declines were noted on day 6. According to Duncan's post-hoc test, these treatments were significantly better than the control, which showed poor firmness-controlling capacity. The improved firmness is likely attributed to a well-balanced combination of biopolymers, which slows down the natural ripening process by limiting water evaporation and oxygen permeability, thereby preserving the structural integrity of the fruit's cell walls and delaying softening (Olivas & Barbosa-Cánovas, 2005).

However, treatments 1 and 6 showed noticeably low firmness, performing similarly to the control ($p > 0.05$), despite better weight loss control. This is because these formulations likely trap too much moisture and CO2, causing anaerobic fermentation inside the fruit surface. This can create a microenvironment that promotes cell wall degradation, leading to tissue softening and accelerated loss of firmness (Santos et al., 2022). Treatments 3 and 4 also showed a sharp drop in firmness initially, but by day 6, they recovered from the falling rate ($p < 0.05$). A similar

incident has been explained by Guerreiro et al. (2015), stating that some coatings applied wet may initially soften the fruit surface due to moisture or mechanical stress. It is because, as the coating dries, cures, and stabilizes, it forms a stronger, more effective barrier that better maintains firmness over time.

3.3.3. Total soluble solids

TSS primarily reflects the sugar content in strawberries and directly correlates with sweetness and consumer quality perception. Postharvest, strawberries experience a decline in TSS due to ongoing respiration, which consumes sugars as an energy source. Two-way ANOVA (Supplementary Table 2) demonstrated that storage time and coatings, as well as their interactions, significantly influence TSS value, consistent with González-Cuello et al. (2024).

A noticeable decline in TSS was observed in all groups (Fig. 7A–B), with control values decreasing from 6.27°Brix on day 2 to 1.17°Brix on day 6 ($p < 0.05$). This rapid sugar degradation is due to moisture loss and cell breakdown, causing enhanced respiration (Nguyen & Nguyen, 2021). Interestingly, treatment 6 showed TSS retention lower than the control, with a day 6 value of 1.03°Brix ($p > 0.05$), which may be attributed to poor barrier formation due to inadequate film integrity or surface incompatibility with the fruit (Garcia et al., 2012).

In contrast, treatments 4, 7, and 3 maintained significantly higher TSS, with treatment 4 exhibiting the highest retention (4.27°Brix) at the end of storage. These coatings' superior performance in moisture and firmness retention suggests reduced cell wall degradation, metabolic activity, and spoilage, helping preserve TSS.

Interestingly, treatments 5 and 9 maintained a steady TSS level during the first two days of storage, while other treatments showed sharp declines. This early stability likely results from an effective initial barrier formed by the coatings, which reduced respiration and moisture loss. However, as storage progressed, their barrier function weakened, allowing metabolic activity to resume and leading to TSS decline ($p < 0.05$) (Mulla et al., 2025).

3.3.4. pH

Fig. 7C presents the pH values of strawberries treated with various edible coatings and controls over 6 days of storage. Storage time, coating

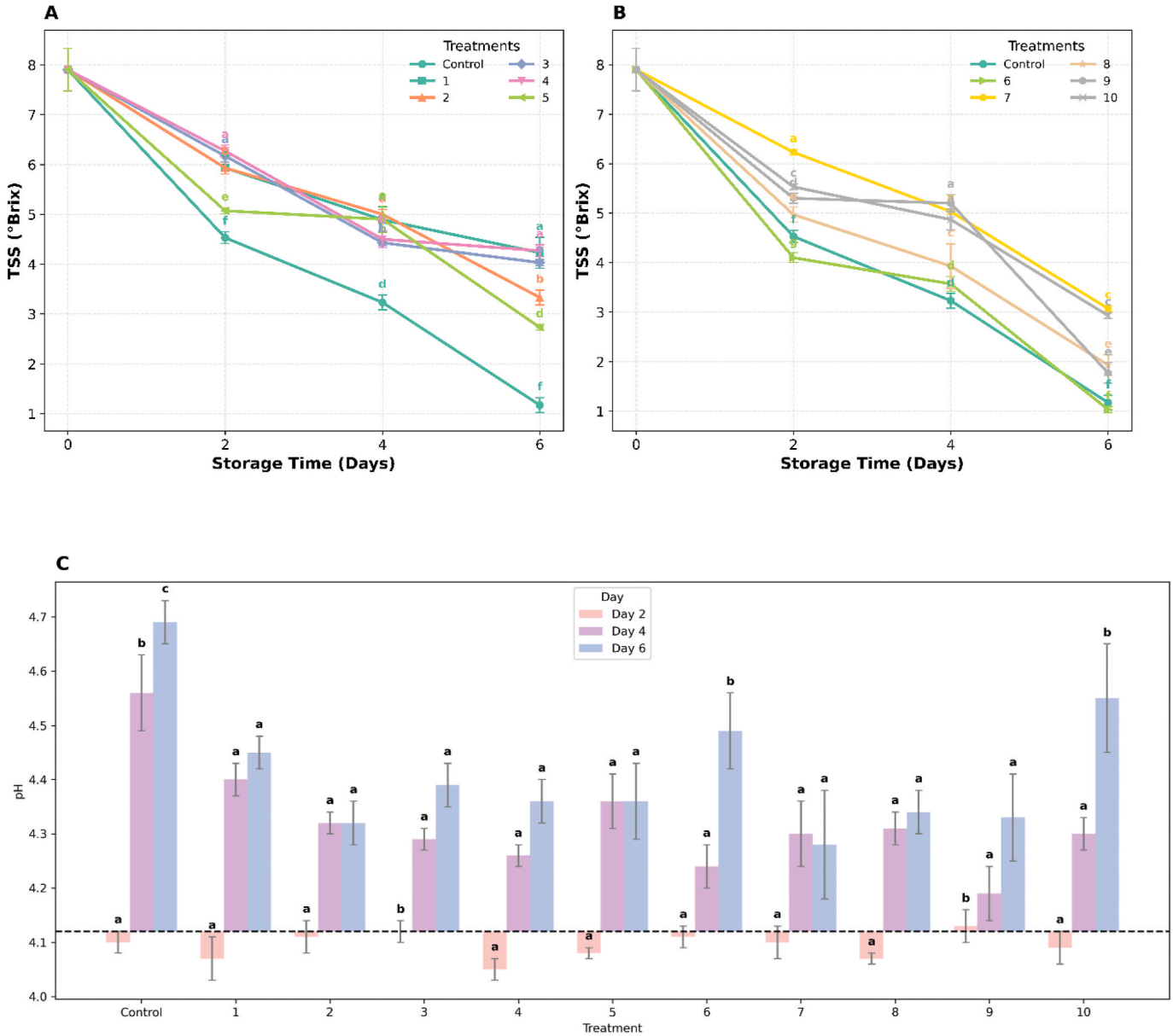


Fig. 7. Effect of different treatments on the TSS (A–B) and pH (C) of strawberries during 6 days of storage. Data are presented as mean ± standard deviation.

treatment, and their interaction showed a statistically significant effect on the pH value of the samples (Supplementary Table 2). Maintaining a lower pH is advantageous for strawberry preservation, as it inhibits the growth of spoilage microorganisms and slows enzymatic browning (Holcroft & Kader, 1999).

In the early stages of storage (days 2 and 4), most treatments showed no significant variation in pH. However, by day 6, the control group displayed a notably higher pH value (4.69). According to Mgaya-Kilima et al. (2014), this is due to the degradation of Vitamin C and lack of antioxidants, allowing alkaline byproducts to accumulate. In contrast, treatments 2, 4, 5, 7, and 9 maintained lower pH values (4.28–4.39), indicating a more acidic environment. According to Barrazueta-Rojas et al. (2018), edible coatings' semi-permeable barrier reduces oxygen permeability and slows down respiration, which in turn helps preserve organic acids and prevent the oxidative degradation of ascorbic acid, both of which are crucial for maintaining a more acidic environment. Treatment 7, which utilized a higher percentage of alginate, was particularly effective in retaining a lower pH (4.28 on day 6). This can be explained by the denser polymer network formed by higher alginate concentrations, which more effectively restricts gas exchange and moisture loss, thereby preserving the fruit's natural acidity (Guilbert et al., 1996).

3.3.5. Color

All color parameters (L*, a*, b*, and ΔE) were significantly affected by storage time, coating types, and their interactions (Supplementary Table 3). L*, a*, and b* data collected over storage periods are provided in Supplementary Table 6. These values were used to calculate the total color difference (ΔE), which serves as a comprehensive indicator of visual quality degradation during storage (Fig. 8).

On day 2, all treatments exhibited relatively low ΔE values, except for the control and treatment 6, which recorded values of 9.57 and 9.96, respectively. A similar trend was also observed on day 6, with the control and treatment 6 and 5, showing the highest ΔE values (~19.0, ~22.0, ~20, respectively), indicating significant visual deterioration. These results suggest the lack of effective mechanisms to inhibit pigment loss. This is likely due to ongoing oxidation and enzymatic browning, resulting from poor moisture resistance and the high hydrophilicity of the coatings, which allowed oxygen and moisture to penetrate the fruit surface (Olivas & Barbosa-Cánovas, 2005). Additionally, in the absence of incorporated antioxidants or antimicrobial compounds, these coatings were less effective at inhibiting the activity of polyphenol oxidase and controlling microbial growth, which accelerated the color degradation (Dhall, 2013).

In contrast, treatment 7 consistently lowest ΔE values (~6.0–~9.5). This can be attributed to the presence of effective natural stabilizing factors within its formulation. Such stabilizers help to mitigate oxidative reactions and enzymatic browning by maintaining a more stable pH and scavenging free radicals, thus reducing the activity of polyphenol oxidase and other enzymes responsible for color changes (Caivano & del Pilar Buera, 2012). This protective effect slows down the breakdown of anthocyanins and other pigments, resulting in less visual deterioration over time.

3.3.6. Total phenolic content

TPC results are presented in Table 3. Phenolic compounds are integral to the antioxidant capacity of strawberries and are distributed across the skin, pulp, seeds, and pedicle. Their concentrations are influenced by temperature, light, enzymatic activity, fruit maturity, and storage conditions (Cosme et al., 2018). Both the storage duration and coating type, as well as their interaction, significantly affected TPC (Supplementary Table 4), consistent with findings by Qaderi et al. (2022).

The data reveal a complex dynamic in phenolic compounds' behavior across treatments. On day 2, the control exhibited the highest TPC (818.81 μg GAE/g FW), followed by treatment 1 (809.67 μg GAE/g FW, $p < 0.05$) and treatment 9 (721.49 μg GAE/g FW, $p < 0.05$). A similar pattern was also observed on day 4, with the control and treatment 6 maintaining the highest TPC (1155.34 μg GAE/g FW and 1130.21 μg GAE/g FW, respectively, $p < 0.05$). This initial elevation in uncoated fruits likely results from stress-induced phenolic biosynthesis and membrane permeabilization caused by postharvest oxidative stress, which activates phenolic biosynthetic pathways (phenylalanine ammonia-lyase, PAL) and releases membrane-bound phenolics, temporarily increasing extractable phenolic content despite overall tissue degradation (Wang et al., 2024).

The initially lower TPC values in coated treatments can be attributed to metabolic suppression. Coatings create a modified atmosphere that reduces respiratory metabolism and phenolic biosynthesis, while simultaneously maintaining cellular integrity and preventing the release of bound phenolics that would otherwise contribute to higher extractable TPC values. By day 6, coated treatments 4, 5, and 7 showed significantly higher TPC than the control, with treatment 4 retaining the highest TPC value (1119.98 μg GAE/g FW). This reversal indicates that

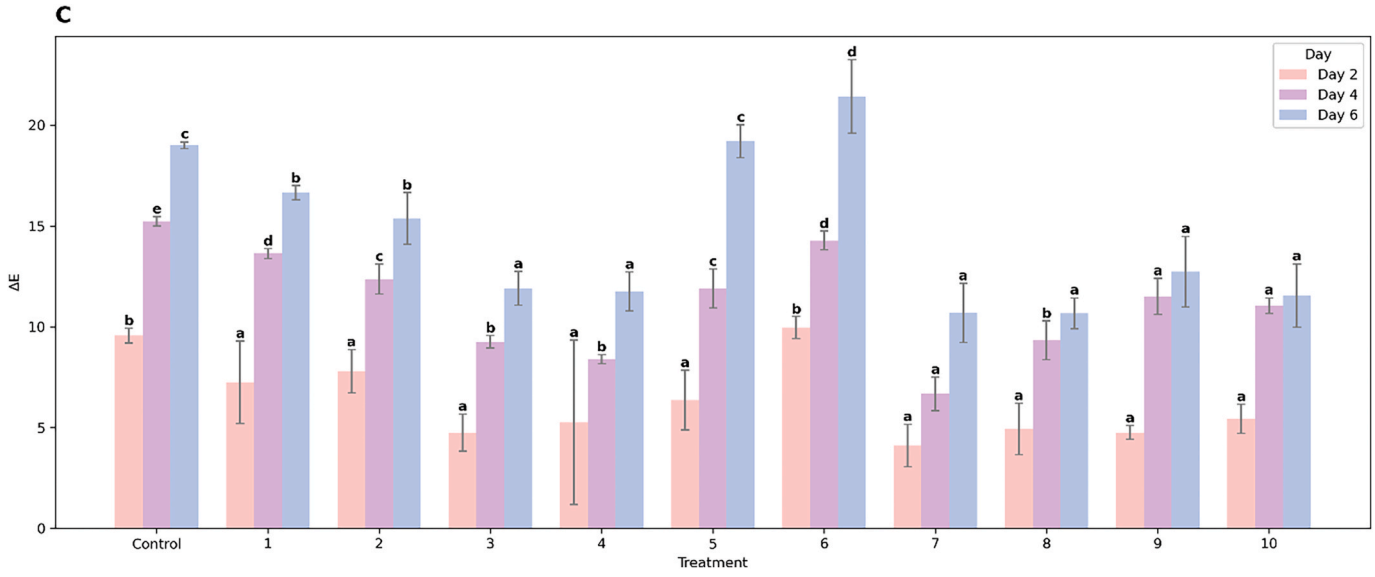


Fig. 8. Color difference (ΔE) of coated and uncoated strawberries during shelf-life evaluation.

Table 3
TPC (μg GAE/g FW) and DPPH (% inhibition) of strawberries during storage.

Treatments	TPC			DPPH		
	2 days	4 days	6 days	2 days	4 days	6 days
Control	818.81 ± 2.29^{a}	1155.34 ± 4.70^{a}	522.88 ± 9.02^{h}	82.43 ± 0.83^{f}	49.43 ± 1.61^{f}	59.74 ± 1.24^{g}
1	809.67 ± 2.33^{b}	962.03 ± 3.38^{e}	777.82 ± 3.26^{f}	79.54 ± 0.49^{g}	76.10 ± 1.18^{c}	79.1 ± 0.41^{c}
2	434.18 ± 3.35^{g}	548.46 ± 6.84^{j}	869.02 ± 3.95^{e}	90.45 ± 0.53^{c}	74.36 ± 1.06^{d}	79.71 ± 0.43^{c}
3	361.01 ± 5.29^{j}	971.86 ± 6.10^{d}	1082.69 ± 4.60^{b}	91.76 ± 0.34^{b}	86.47 ± 0.90^{b}	87.07 ± 0.33^{a}
4	411.30 ± 9.29^{h}	461.35 ± 3.80^{k}	1119.98 ± 5.48^{a}	93.45 ± 0.16^{a}	54.83 ± 1.99^{e}	88.27 ± 0.66^{a}
5	567.46 ± 1.98^{e}	1050.75 ± 3.21^{c}	1116.19 ± 5.22^{a}	89.36 ± 0.66^{d}	84.68 ± 2.31^{b}	70.7 ± 0.16^{e}
6	404.13 ± 7.23^{h}	1130.21 ± 6.71^{b}	636.64 ± 2.01^{g}	85.11 ± 0.59^{e}	76.11 ± 2.57^{c}	56.74 ± 1.59^{h}
7	370.04 ± 2.72^{i}	684.45 ± 2.90^{i}	1028.81 ± 6.77^{c}	91.65 ± 0.49^{b}	88.44 ± 2.47^{b}	85.87 ± 0.81^{b}
8	461.74 ± 4.80^{f}	740.89 ± 4.87^{h}	1028.8 ± 3.83^{c}	90.18 ± 0.33^{c}	78.46 ± 1.47^{c}	72.51 ± 1.16^{d}
9	721.49 ± 4.77^{c}	839.14 ± 8.04^{f}	780.15 ± 3.63^{f}	92.04 ± 0.25^{b}	92.09 ± 1.06^{a}	63.72 ± 1.75^{f}
10	599.40 ± 3.71^{d}	817.44 ± 4.87^{g}	910.06 ± 2.80^{d}	94.0 ± 0.25^{a}	85.27 ± 1.30^{b}	86.31 ± 0.34^{a}

*Data are mean ± standard deviation (SD). Mean values with different letters in a column indicate a significant difference at $p < 0.05$; TPC = Total phenolic content; DPPH = 2,2-diphenyl-1-picrylhydrazyl.

effective coatings with higher alginate prevented phenolic degradation by inhibiting polyphenol oxidase (PPO) activity through limited oxygen availability, reducing leaching losses of water-soluble phenolics, and maintaining biosynthetic capacity through preserved cellular integrity (Nguyen & Nguyen, 2021). By reducing oxidative stress and inhibiting the activity of oxidative enzymes like polyphenol oxidase, the coatings help maintain the integrity of phenolic compounds, resulting in higher TPC values at later stages of storage. This is further supported by the lower weight loss and improved firmness in these treatments, which likely contributed to better retention of phenolic compounds.

3.3.7. Antioxidant activity

Table 3 displays the antioxidant activity of strawberries as measured by the DPPH radical scavenging assay. Higher DPPH values are associated with better retention of color, flavor, and nutritional quality, as antioxidants counteract oxidative damage (Milosavljević et al., 2022). Storage period, coating type, and their interaction significantly influenced DPPH activity (Supplementary Table 4), aligning with findings by De Bruno et al. (2023).

The control group showed a marked decline, dropping from 82.43 % on day 2–49.43 % on day 4 ($p < 0.05$), followed by a significant recovery to 59.74 % on day 6. This pattern reflects oxidative degradation in uncoated fruits, with the subsequent increase possibly resulting from stress-induced synthesis of phenolic compounds or activation of endogenous antioxidant enzymes (Fadda et al., 2015). Treatments 6 and 1 also showed consistently poor antioxidant preservation capacity throughout the storage period ($p < 0.05$), indicating insufficient barrier properties or potential incompatibility with the fruit surface (Dhall, 2013).

Among the treatments, treatment 10 exhibited the most stable performance, maintaining high DPPH values from 94.0 % on day 2–86.31 % on day 6 ($p > 0.05$), indicating strong preservation of antioxidant compounds. This suggests that the applied coating effectively reduced oxidative stress, thereby minimizing antioxidant degradation (Dhall, 2013). Treatments 3 and 7 exhibited similar results, with no significant differences in DPPH values throughout storage. Notably, these treatments also excelled in maintaining other crucial physicochemical attributes. Treatment 7, in particular, showed superior moisture retention and firmness, supporting the hypothesis that preserving these parameters helps prevent cellular disruption and the subsequent release of oxidative enzymes responsible for antioxidant loss (Olivas & Barbosa-Cánovas, 2005).

The observed trends in TPC and DPPH highlight the interplay between phenolic compound preservation and oxidative stress mitigation during storage. While uncoated strawberries initially show elevated TPC due to stress-induced phenolic synthesis and membrane disruption (Wang et al., 2024), their respective DPPH values dropped significantly. In contrast, coated fruits exhibit more stable and sustained TPC and antioxidant activity, attributable to coatings' ability to create a controlled atmosphere that limits oxygen exposure, reduces enzymatic oxidation, and prevents phenolic leaching (Nguyen & Nguyen, 2021). This correlation between maintained moisture, firmness, and antioxidant retention underscores the importance of coatings in preserving cellular integrity, thus enhancing both phenolic stability and overall antioxidant capacity during postharvest storage.

3.4. Correlation among physicochemical and bioactive properties

The correlation matrix (Fig. 9) reveals a meaningful relationship between the physicochemical and biological properties of coated strawberries. This highlights the interconnected nature of post-harvest deterioration.

Firmness emerged as a sensitive integrator of overall quality. It showed a strong negative correlation with ΔE ($r = -0.91$) and pH ($r = -0.76$), a positive correlation with TSS ($r = 0.82$) and DPPH activity ($r = 0.70$), and a moderately negative correlation with TPC ($r = -0.50$). Collectively, these patterns indicate that as fruit softens, visual quality declines, acidity drops, and cellular/biochemical integrity deteriorates (Day et al., 2009; Rodríguez et al., 2014; Zhang et al., 2022). Coatings that failed to retain texture, such as treatment 6, also suffered higher ΔE and pH, lower TSS, and inconsistent antioxidant responses. In contrast, treatments that maintained good firmness showed better sweetness (TSS), visual quality, and more stable antioxidant capacity, aligning with prior reports (Day et al., 2009; Rodríguez et al., 2014).

Weight loss was also found to be strongly associated with multiple strawberry attributes, showing a high negative correlation with firmness ($r = -0.86$), TSS ($r = -0.90$), and DPPH ($r = -0.65$), and a positive correlation with ΔE ($r = 0.83$). This suggests that moisture loss is highly associated with softening, sugar depletion, and oxidative release of phenolics and antioxidants from damaged tissues (Dhall, 2013; Lowder & Osburn, 2010). Treatment 6, for instance, exhibited 28.95 % weight loss by day 6, accompanied by severe color change and rising pH. Conversely, treatments 7 and 4, both of which limited weight loss to below 17 %, retained more favorable quality attributes. ΔE positively correlated with pH ($r = 0.76$), TPC ($r = 0.46$), and DPPH ($r = -0.29$), further supporting the notion that oxidation and pigment degradation coincide with acid loss and phenolic release (Dhall, 2013).

TPC and DPPH showed only a weak negative correlation ($r \approx -0.20$), indicating that the total pool of phenolics is a poor single-predictor of radical-scavenging capacity. This decoupling is well documented in fruit systems and arises from (i) compositional effects, antioxidant activity depends on phenolic subclass and substitution pattern (e.g., ortho-dihydroxyl groups, degree of polymerization) rather than total concentration (Milosavljević et al., 2022), (ii) assay chemistry, the Folin–Ciocalteu reagent responds to many non-phenolic reductants (ascorbate, some sugars, amino acids), whereas DPPH primarily reflects

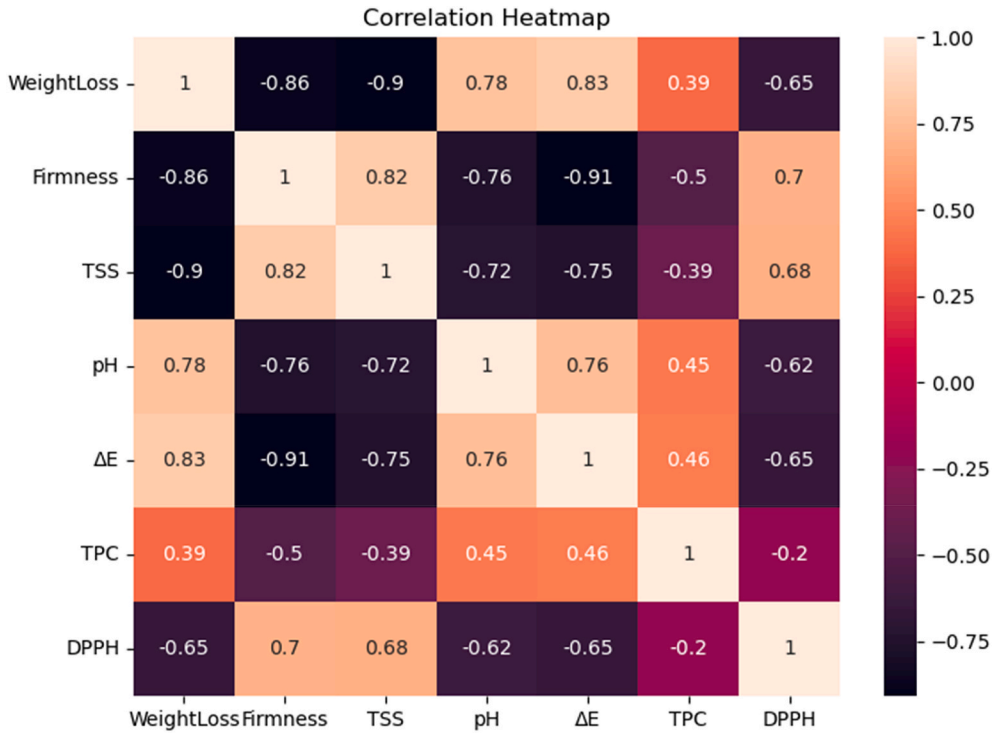


Fig. 9. Pearson correlation matrix among physicochemical quality parameters of strawberries.

the electron/hydrogen-donating ability of specific phenolics in an organic medium (Singleton et al., 1999). Consistent with prior strawberry studies, our study also observed cases where DPPH was relatively well preserved despite moderate TPC (e.g., treatments 3, 7, 10), suggesting that coatings may favor retention of more redox-active phenolics and ascorbate and/or suppress oxidative enzymes, rather than simply maximizing the bulk phenolic pool (De Bruno et al., 2023; Wani et al., 2021). Together, these results and the literature indicate that "antioxidant activity" reflects phenolic quality and redox state more than total content; consequently, TPC should be interpreted alongside activity assays (e.g., DPPH, ABTS, FRAP) when evaluating coating efficacy.

3.5. Predictive modeling of storage conditions and coating compounds

RF, SVR, and XGB machine-learning models were trained using experimental data gathered during strawberry storage. The models iteratively adjusted their hyperparameters during training in order to reduce the Root Mean Squared Error (RMSE) loss function. Additional metrics, such as the coefficient of determination (R^2) and mean absolute error (MAE), were used to further assess the model's performance.

The SVR model outperformed both the RF and XGB models across all evaluation metrics, achieving the lowest MAE, MSE, and RMSE in both the cross-validation and evaluation datasets (Table 4), indicating its strong ability to generalize to unseen data. SVR was particularly effective in predicting weight loss, TSS, pH, and ΔE, and it also achieved the lowest test error for firmness, surpassing other models in both accuracy and consistency (Supplementary Table 7). The superior performance of SVR is likely due to its kernel-based approach, which effectively models the nonlinear and complex relationships in the small strawberry storage dataset, as observed in studies such as (Chen et al., 2022). In contrast, the RF model showed a slightly better RMSE than XGB during cross-validation a evaluation. XGB exhibited similar performance, showing good generalization ability. It is because they both use decision trees; hence, it generalizes well when the training data is sufficient (Li et al., 2021).

Table 4
Machine learning model scoring metrics.

Model Type	Loss Function	CV Train	CV Test	Evaluation
RF	R^2	98.74 %	93.90 %	98.59 %
	MAE	1.52	3.81	1.46
	RMSE	2.36	5.48	3.49
SVR	R^2	99.10 %	95.17 %	98.79 %
	MAE	0.34	0.79	0.44
	RMSE	0.50	1.39	1.21
XGB	R^2	98.33 %	95.28 %	97.61 %
	MAE	2.55	4.03	2.85
	RMSE	3.45	5.29	7.28

*RF= Random Forest; SVR= Support Vector Regression; XGB = eXtreme Gradient Boosting; RMSE = Root Mean Squared Error; MAE = Mean Absolute Error; R^2 = coefficient of determination.

These behaviors were also reflected in the R^2 plots (Supplementary Figs. 2–5), where SVR demonstrated the strongest linear alignment between predicted and actual values across all five output variables. Firmness and weight loss exhibited the highest explained variance across models, while ΔE and pH showed slight fluctuation.

A learning curve (Fig. 10) shows a steady drop in RMSE with training size and a small, stable train validation gap for SVR, evidence of no severe overfitting but some expected variance. The curve doesn't fully plateau and flattens after ~60 groups, implying that more distinct data could still help improve performance.

A 5 x 2CV paired *t*-test was used to compare the models' relative performance in a pairwise comparison. The dataset was divided into two equal-sized subsets using this method, and the model was trained on one subset and tested on another for every iteration. The train and test splits were then switched, and the procedure was repeated. A paired *t*-test was then used to compare the performance metrics derived from these iterations to determine statistical significance. This method addressed limitations found in traditional paired t-tests and k-fold cross-validated t-tests by reducing dependency between training and testing samples and providing a more reliable estimate of performance differences. The result from the 5 x 2CV paired *t*-test indicates no significant performance differences among the models.

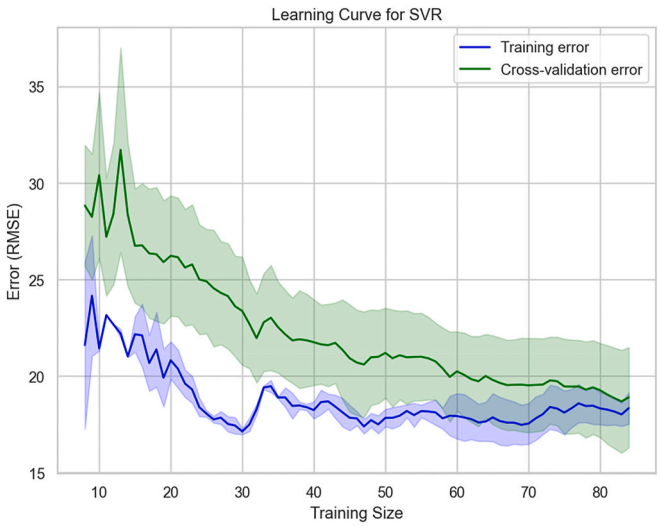


Fig. 10. SVR learning curve with 5-fold CV. Mean ± SD RMSE versus training size.

The model ranks SVR, RF, and XGB in descending order of performance according to error metrics, RMSE, and MAE. The models, when compared using the error metric R^2, rank as SVR (98.79 %), RF (98.59 %), and XGB (97.61 %).

3.6. Factor effects and variable importance

The construction of machine learning models enabled the estimation of the relative importance of each input variable through permutation feature importance. This method assesses the significance of a feature by randomly shuffling its values and measuring the resulting increase in prediction error. A larger increase in error indicates greater feature importance (Casalicchio et al., 2019). The results (Fig. 11) clearly show that storage duration (days) has the most influential effect on all the physicochemical attributes across all models, aligning with the findings of Farooq et al. (2020), reporting that extended storage leads to deterioration in quality parameters like firmness, weight loss, and soluble solids content, primarily due to the breakdown of cell structures and ongoing metabolic activities.

Following storage duration, the coating composition was identified as the second most important factor affecting strawberry quality. Alginate emerged as the most influential component in the edible coating formulation. Alginate's superior performance is attributed to its ability to form a dense gel matrix that restricts moisture loss and controls gas exchange. This aligns with findings by Muley and Singhal (2020), who noted that alginate coatings significantly reduce the water vapor transmission rate and delay ripening in fruits.

Guar gum and pectin also contributed to the coating's effectiveness, but in a more complementary role. Guar gum, known for its water-holding capacity and emulsifying properties, and pectin, with its gelling and antioxidant properties, helped enhance the flexibility and durability of the coatings. Pectin was found to be a more important parameter (Fig. 11) for both the SVR and XGB models, whereas the RF model identified guar gum as more important. These findings are consistent with the work of Gull et al. (2024), who demonstrated that the overall properties of the gel matrix were improved by combining with other polymers.

3.7. Optimization of strawberries' physicochemical attributes

By integrating the three trained models with the DE and PSO optimizers, the coating formulation and storage period that achieves the highest postharvest quality of strawberries were determined (Table 5). Because each learner captures different nonlinearities, their optima need

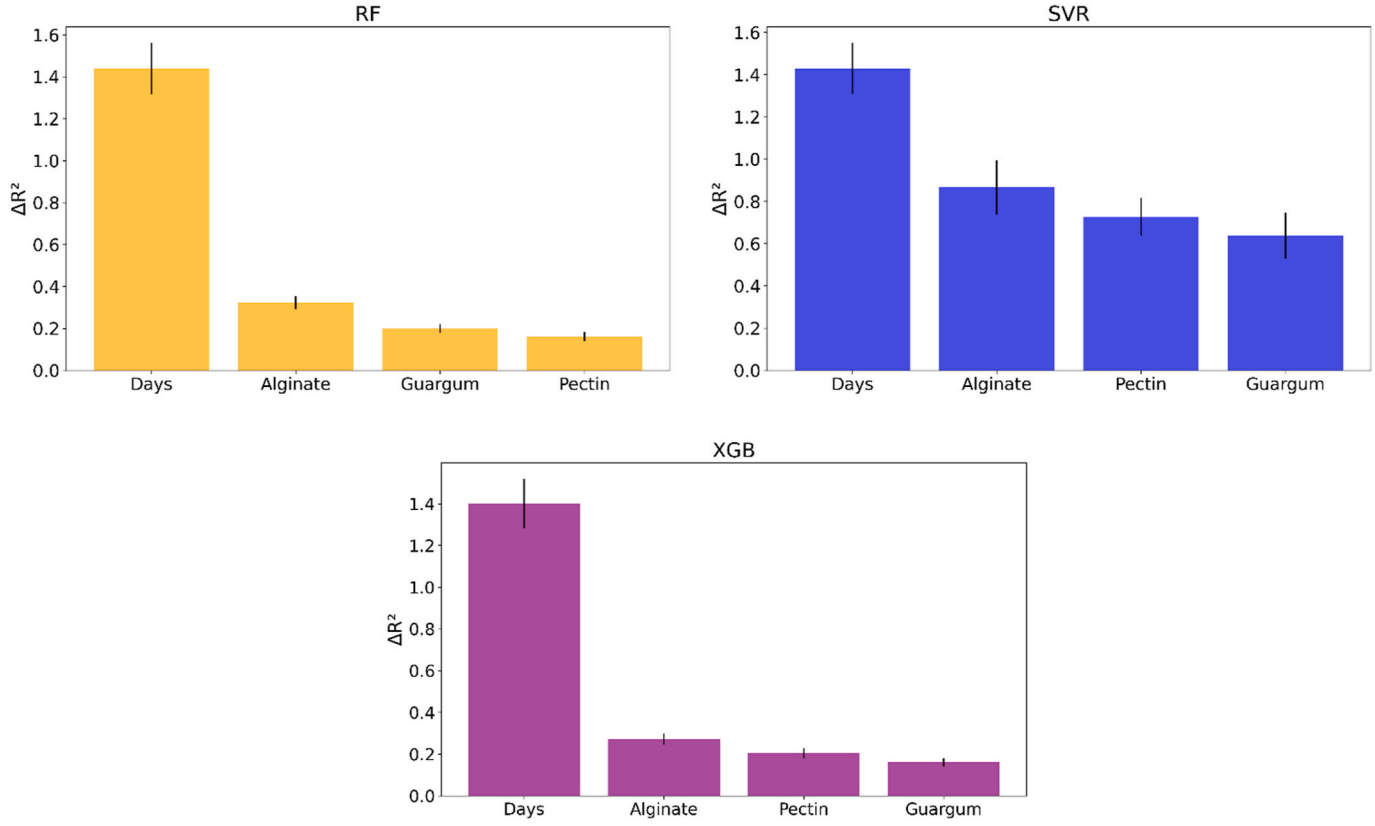


Fig. 11. The relative importance of model input features.

Table 5
Machine learning facilitated optimization of the strawberries' physicochemical properties.

Model	Optimization Method	Time	Alginate	Guar Gum	Pectin	Weight Loss (%)	Firmness (N)	TSS	pH	ΔE
RF	DE	6	0.56	0.33	2.11	7.00	586.34	5.44	4.08	4.32
	PSO	3.46	1.37	1.52	0.11	7.78	649.97	5.64	4.07	6.18
SVR	DE	2.78	1.80	0.60	0.60	8.93	696.08	6.53	4.10	3.29
	PSO	3.80	1.83	0.63	0.54	9.64	678.96	6.22	4.19	5.54
XGB	DE	6	0.97	0.55	1.48	6.89	570.45	6.11	4.07	5.75
	PSO	2.82	0.89	0.23	1.88	6.82	643.12	6.13	4.09	4.55

RF= Random Forest; SVR = Support Vector Regressor, XGB = Extream Gradient Boosting, DE = Differential Evolution, PSO = Particle Swarm Optimization.

not coincide. Differential Evolution tended to find the 6-day optima, whereas PSO converged to ~2.8–3.8 days with occasional metric gains (e.g., XGB–PSO ΔE 4.55, firmness 643 N) but shorter shelf-life. Our study, therefore, interprets (i) the pattern across models/optimizers and (ii) the trade-off between shelf-life and quality.

The optimization revealed two practical families of coatings. According to the findings from Table 5, pectin-rich blends (pectin ≈1.48–2.11 % with alginate 0.56–0.97 % and guar 0.23–0.55 %) produced lower weight loss and restrained color change at the longest horizons: weight loss 6.82–7.00 %, ΔE 4.32–5.75, firmness 570.45–643.12 N, with up to 6 days at 25 °C. In contrast, alginate-forward, balanced blends (alginate ≈1.80–1.83 % with ≈0.6 % each guar and pectin) achieved the highest firmness (674.97–696.08 N) but only at shorter times (2.78–3.80 days), and with higher weight loss (7.78–9.64 %) and variable color (ΔE 3.29–6.18). This pattern is mechanistically coherent: higher pectin improves film continuity and moisture/gas barriers (less mass transfer, slower color drift) (Syarifuddin et al., 2025), whereas higher alginate strengthens the gel network that supports texture early, but does not limit water loss as effectively (Sotera et al., 2021). Notably, a guar gum-tilted point (RF–PSO 1.37/1.52/0.11) delivered only 3.46 days with ΔE 6.18, suggesting excess guar adds viscosity without proportional barrier or mechanical gains. Given the aim to maximize time at 25 °C (298.15K) with acceptable quality, RF–DE (0.56/0.33/2.11 %) finds the best optimal: 6 days, weight loss 7.00 %, ΔE 4.32, firmness 586.34 N, and TSS 5.44.

4. Conclusion

Our research demonstrates the effectiveness of data augmentation in generating high-quality data by utilizing inherent correlations within the dataset, addressing the issue of data scarcity. The study further highlights the precision of three machine learning models (RF, XGB, and SVR) as reliable multi-objective optimization tools, providing comparable performance and predictions.

The data analysis revealed that alginate, guar gum, and pectin-based edible coatings significantly improve the postharvest quality of strawberries. Correlation analysis identified strong interdependencies between key quality parameters such as firmness, weight loss, ΔE, TSS, DPPH, and TPC, demonstrating their co-evolution during postharvest senescence. Through ML-based optimization, two formulation families emerged: pectin-rich coatings extended shelf life by reducing weight loss and color change, while higher alginate content improved firmness. The optimal formulation, consisting of 0.56 % alginate, 0.33 % guar gum, and 2.11 % pectin, demonstrated a 7 % weight loss, firmness of 586.34 N, TSS of 5.44, pH of 4.08, and ΔE of 4.32 over 6 days at 25 °C (298.15K).

5. Limitations and future directions

This study is based on several assumptions: the coating polymers and storage duration influence strawberry quality; machine learning can accurately predict physicochemical changes; and these predictions align closely with experimental results. While the findings provide useful insights for optimizing postharvest handling and reducing waste, the study is limited to a single cultivar and set of storage and coating conditions. Furthermore, evaluation was restricted to core quality markers; future work should quantify respiration rate and microbial decay and profile coating physicochemical properties (particle size/PDI, zeta potential, rheology, contact angle, WVTR/OTR), alongside sensory testing.

The experiments were conducted at 25 °C (298.15K) to emulate warm-chain retail conditions common in tropical markets. Because cooling markedly reduces respiration, transpiration, and enzymatic activity, these findings should not be linearly extrapolated to standard refrigerated storage (0–4 °C). Under refrigeration, the absolute magnitude of improvements in weight loss, firmness, color, and acids is expected to be smaller. However, the relative ranking of effective vs. ineffective formulations may be retained. Future work will therefore validate the best two to three blends at 0–2 °C (273.15–275.15K) and 4 °C (277.15K) over extended storage (≥10–14 days). Future research ought to evaluate the coatings' microbial stability, test them on various strawberry cultivars, and validate the models in a range of storage scenarios, including actual supply chains. Enhancing shelf life and quality will require investigating novel biopolymer blends, bioactive ingredients, and sensory evaluation. Other perishable produce could also benefit from this ML-guided strategy to lower postharvest losses.

CRediT authorship contribution statement

Saklain Niam: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Iftekhar Ahmad:** Writing – review & editing, Supervision, Conceptualization. **Md Abu Rayhan:** Writing – review & editing. **Sajid Mahmood:** Writing – review & editing. **Parvej Hasan Jon:** Writing – review & editing. **Md Monir Ahmed:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.lwt.2025.118548.

Data availability

The datasets, statistical analysis, and machine learning code used in this study are available on GitHub at: https://github.com/saqlineniam/Strawberry_Edible_coating.

References

Abookleesh, F., Zubair, M., & Ullah, A. (2025). Leveraging machine learning for the optimization of reinforced rapeseed protein-gelatin edible coatings for enhanced food preservation. *Chemical Engineering Journal, 512*, Article 162604.

Balic, I., Ejsmentewicz, T., Sanhueza, D., Silva, C., Peredo, T., Olmedo, P., Barros, M., Verdonk, J. C., Paredes, R., & Meneses, C. (2014). Biochemical and physiological study of the firmness of table grape berries. *Postharvest Biology and Technology, 93*, 15–23.

Barrazueta-Rojas, S. G., Falconí, J. F., Navarro-Ojeda, M. N., Oleas-López, J. M., & Mendoza-Zurita, G. X. (2018). Pysicochemical properties and application of edible coatings in strawberry Fragaria x Ananassa) preservation. *Revista Facultad Nacional de Agronomía Medellín, 71*(3), 8631–8641.

Basu, A., Nguyen, A., Betts, N. M., & Lyons, T. J. (2014). Strawberry as a functional food: An evidence-based review. *Critical Reviews in Food Science and Nutrition, 54*(6), 790–806.

Caivano, J. L., & del Pilar Buera, M. (2012). *Color in food: Technological and psychophysical aspects*. CRC Press.

Carole, D. A., Derricka, M. D. R. T., Dé, M. H., Souaibou, A., Cé, M. E., Joë, B. N., Dioril, A. D., Didier, M. B., Charlen, N. I. B., & Emmanuel, K. (2024). Strawberry (Fragaria spp.): Cultivation, production, consumption, and marketing in Cameroon. *Agricultural Sciences, 15*(4), 449–471.

Casalicchio, G., Molnar, C., & Bischl, B. (2019). Visualizing the feature importance for black box models. *Machine learning and knowledge discovery in databases: European conference*. ECML PKDD 2018, Dublin, Ireland, September 10–14, 2018, Proceedings, Part I 18.

Chandra, D., Choi, A. J., Lee, J. S., Lee, J., & Kim, J. G. (2015). Changes in physicochemical and sensory qualities of "Goha" strawberries treated with different conditions of carbon dioxide. *Agricultural Sciences, 6*(3), 325.

Chen, X., Jiang, Z., Tai, Q., Shen, C., Rao, Y., & Zhang, W. (2022). Construction of a photosynthetic rate prediction model for greenhouse strawberries with distributed regulation of light environment. *Mathematical Biosciences and Engineering, 19*, 12774–12791.

Cosme, F., Pinto, T., & Vilela, A. (2018). Phenolic compounds and antioxidant activity in grape juices: A chemical and sensory view. *Beverages, 4*(1), 22.

Cristina, S. (2021). *Differential evolution from scratch in python*. Machine Learning Mastery.

da Silva Rios, D. A., Nakamoto, M. M., Braga, A. R. C., & da Silva, E. M. C. (2022). Food coating using vegetable sources: Importance and industrial potential, gaps of knowledge, current application, and future trends. *Applied Food Research, 2*(1), Article 100073.

Day, L., Seymour, R. B., Pitts, K. F., Konczak, I., & Lundin, L. (2009). Incorporation of functional ingredients into foods. *Trends in Food Science & Technology, 20*(9), 388–395.

De Bruno, A., Gattuso, A., Ritorto, D., Piscopo, A., & Poiana, M. (2023). Effect of edible coating enriched with natural antioxidant extract and bergamot essential oil on the shelf life of strawberries. *Foods, 12*(3), 488.

de Souza, W. F. C., de Lucena, F. A., da Silva, K. G., Martins, L. P., de Castro, R. J. S., & Sato, H. H. (2021). Influence of edible coatings composed of alginate, galactomannans, cashew gum, and gelatin on the shelf-life of grape cultivar 'italia': Physicochemical and bioactive properties. *LWT, 152*, Article 112315.

Dhall, R. (2013). Advances in edible coatings for fresh fruits and vegetables: A review. *Critical Reviews in Food Science and Nutrition, 53*(5), 435–450.

Du, Y., & Li, N. (2024). Towards principled assessment of tabular data synthesis algorithms. *CoRR*.

Fadda, C., Fenu, P. A. M., Usai, G., Caro, A.d., Diez, Y. M., Sanguinetti, A. M., & Piga, A. (2015). Antioxidant activity and sensory changes of strawberry tree fruits during cold storage and shelf life. *Czech Journal of Food Sciences*.

Falcao-Rodrigues, M. M., Moldao-Martins, M., & Beirão-da-Costa, M. L. (2007). DSC as a tool to assess physiological evolution of apples preserved by edibles coatings. *Food Chemistry, 102*(2), 475–480.

Farooq, S., A. Rather, S., Gull, A., Ahmad Ganai, S., Masoodi, F., Mohd Wani, S., & Ganaie, T. A. (2020). Physicochemical and nutraceutical properties of tomato powder as affected by pretreatments, drying methods, and storage period. *International Journal of Food Properties, 23*(1), 797–808.

Garcia, L. C., Pereira, L. M., de Luca Sarantópoulos, C. I., & Hubinger, M. D. (2012). Effect of antimicrobial starch edible coating on shelf-life of fresh strawberries. *Packaging Technology and Science, 25*(7), 413–425.

González-Cuello, R., Parada-Castro, A. L., & Ortega-Toro, R. (2024). Application of a multi-component composite edible coating for the preservation of strawberry fruit. *Journal of Composites Science, 8*(12), 515.

Guerreiro, A. C., Gago, C. M., Faleiro, M. L., Miguel, M. G., & Antunes, M. D. (2015). The effect of alginate-based edible coatings enriched with essential oils constituents on Arbutus unedo L. fresh fruit storage. *Postharvest Biology and Technology, 100*, 226–233.

Guilbert, S., Gontard, N., & Gorris, L. G. (1996). Prolongation of the shelf-life of perishable food products using biodegradable films and coatings. *LWT-Food Science and Technology, 29*(1–2), 10–17.

Gull, S., Ejaz, S., Ali, S., Ali, M. M., Sardar, H., Azam, M., Deng, H., Yousef, A. F., Alrefaei, A. F., & Almutairi, M. H. (2024). Xanthan gum-based edible coating effectively preserve postharvest quality of 'gola'guava fruits by regulating physiological and biochemical processes. *BMC Plant Biology, 24*(1), 450.

Holcroft, D. M., & Kader, A. A. (1999). Controlled atmosphere-induced changes in pH and organic acid metabolism may affect color of stored strawberry fruit. *Postharvest Biology and Technology, 17*(1), 19–32.

Kader, A. A. (2002). *Postharvest technology of horticultural crops, 3311*. University of California Agriculture and Natural Resources.

Khodaei, D., Hamidi-Esfahani, Z., & Rahmati, E. (2021). Effect of edible coatings on the shelf-life of fresh strawberries: A comparative study using TOPSIS-shannon entropy method. *NFS Journal, 23*, 17–23.

Kumar, N., Daniloski, D., D'cunha, N. M., Naumovski, N., & Petkoska, A. T. (2022). Pomegranate peel extract–A natural bioactive addition to novel active edible packaging. *Food Research International, 156*, Article 111378.

Kumar, N., & Neeraj. (2019). Polysaccharide-based component and their relevance in edible film/coating: A review. *Nutrition & Food Science, 49*(5), 793–823.

Kumar, N., Pratibha, T., Petkoska, A., Khojah, E., Sami, R., & Al-Mushhin, A. A. (2021). Chitosan edible films enhanced with pomegranate peel extract: Study on physical, biological, thermal, and barrier properties. *Materials, 14*(12), 3305.

Kumar, N., Pratibha, Upadhyay, A., Trajkovska Petkoska, A., Gniewosz, M., & Kieliszek, M. (2023). Extending the shelf life of mango (Mangifera indica L.) fruits by using edible coating based on xanthan gum and pomegranate peel extract. *Journal of Food Measurement and Characterization, 17*(2), 1300–1308.

Kumar, S., Reddy, A. R. L., Basumatary, I. B., Nayak, A., Dutta, D., Konwar, J., Purkayastha, M. D., & Mukherjee, A. (2023). Recent progress in pectin extraction and their applications in developing films and coatings for sustainable food packaging: A review. *International Journal of Biological Macromolecules, 239*, Article 124281.

Kumarihami, H. P. C., Kumar, N., Pratibha, Petkoska, A. T., & Neeraj. (2023). Effect of natural materials on thermal properties of food packaging film: An overview. *Natural Materials for Food Packaging Application*, 255–274.

Li, J., Liu, H., Yang, Z., & Han, L. (2021). A credit risk model with small sample data based on G-XGBoost. *Applied Artificial Intelligence, 35*(15), 1550–1566.

Lowder, A. C., & Osburn, W. (2010). Inclusion of blended lipid solutions as functional ingredients to alter the fatty acid profile of beef patties. *Journal of Food Science, 75*(7), S355–S364.

Makhathini, N., Kumar, N., & Fawole, O. A. (2025). Enhancing circular bioeconomy: Alginate-cellulose nanofibre films/coatings functionalized with encapsulated pomegranate peel extract for postharvest preservation of pomegranate arils. *International Journal of Biological Macromolecules, 309*, Article 142848.

Mdenye, B. B., Kinama, J. M., Olubayo, F. M., Kivuva, B. M., & Muthomi, J. W. (2016). Effect of storage methods on carbohydrate and moisture of cassava planting materials. *Journal of Agricultural Science, 8*(12), 100–111.

Menichetti, G., Ravandi, B., Mozaffarian, D., & Barabási, A.-L. (2023). Machine learning prediction of the degree of food processing. *Nature Communications, 14*(1), 2312.

Mgaya-Kilima, B., Remberg, S. F., Chove, B. E., & Wicklund, T. (2014). Influence of storage temperature and time on the physicochemical and bioactive properties of roselle-fruit juice blends in plastic bottle. *Food Science and Nutrition, 2*(2), 181–191.

Milosavljević, D. M., Maksimović, V. M., Milivojević, J. M., Nakarada, Đ. J., Mojović, M. D., & Dragišić Maksimović, J. J. (2022). Rich in phenolics—strong antioxidant fruit? Comparative study of 25 strawberry cultivars. *Plants, 11*(24), 3566.

Moghadas, H. C., Smith, J. S., & Tahergorabi, R. (2025). Recent advances in the application of edible coatings for shelf-life extension of strawberries: A review. *Food and Bioprocess Technology, 18*(2), 1079–1103.

Mudgil, D., Barak, S., & Khatkar, B. S. (2014). Guar gum: Processing, properties and food applications—a review. *Journal of Food Science and Technology, 51*(3), 409–418.

Muley, A. B., & Singhal, R. S. (2020). Extension of postharvest shelf life of strawberries (Fragaria ananassa) using a coating of chitosan-whey protein isolate conjugate. *Food Chemistry, 329*, Article 127213.

Mulla, M. F. Z., Shonte, T., & Pathania, S. (2025). Quality parameters and shelf life of strawberry (cv. Centenary) fruits as affected by active modified atmosphere packaging. *International Journal of Food Science and Technology*, Article vvaf010.

Nair, M. S., Tomar, M., Punia, S., Kukula-Koch, W., & Kumar, M. (2020). Enhancing the functionality of chitosan-and alginate-based active edible coatings/films for the preservation of fruits and vegetables: A review. *International Journal of Biological Macromolecules, 164*, 304–320.

Nguyen, V. T., & Nguyen, H. V. (2021). Postharvest quality of strawberry (Fragaria× ananassa duch.) coated with calcium and nano-chitosan as affected by different storage temperatures. *Journal of Horticulture and Postharvest Research, 4*(4), 413–426.

Nishizawa, T. (2021). Current status and future prospect of strawberry production in east Asia and Southeast Asia. *IX international strawberry symposium*.

Olivas, G., & Barbosa-Cánovas, G. (2005). Edible coatings for fresh-cut fruits. *Critical Reviews in Food Science and Nutrition, 45*(7–8), 657–670.

Prasad, J., Kumar, N., Jaiswal, R., Yadav, A., Sharma, S. P., Fawole, O. A., & Kaushik, N. (2025). Biopolymer based composite packaging: A sustainable approach for fruits and vegetables preservation. *Applied Food Research*, Article 101211.

Priyadarshi, R., Jayakumar, A., de Souza, C. K., Rhim, J. W., & Kim, J. T. (2024). Advances in strawberry postharvest preservation and packaging: A comprehensive review. *Comprehensive Reviews in Food Science and Food Safety, 23*(4), Article e13417.

Qaderi, R., Mezzetti, B., Capocasa, F., & Mazzoni, L. (2022). Influence of different storage conditions on strawberry fruit nutritional quality. *XXXI International Horticultural Congress (IHC2022): International symposium on postharvest technologies to reduce food losses 1364*.

Qiu, W., Zhu, F., Hao, T., Wang, M., & Huang, R. (2025). MBLSTM is a contextual interaction refined method for time series prediction. *Scientific Reports, 15*(1), Article 18563.

Rabasco-Vílchez, L., Jiménez-Jiménez, F., Possas, A., Brunner, M., Fleck, C., & Pérez-Rodríguez, F. (2024). Evaluating the shelf life of strawberries using a portable Vis-NIR spectrophotometer and a Reflectance Quality Index (RQI). *Postharvest Biology and Technology, 218*, Article 113189.

Rahman, H., Ahmad, I., Jon, P. H., Salam, A., & Rabbi, M. F. (2024). Automated detection of selected tea leaf diseases in Bangladesh with convolutional neural network. *Scientific Reports, 14*(1), Article 14097.

Rasera, G. B., Hilkner, M. H., de Alencar, S. M., & de Castro, R. J. S. (2019). Biologically active compounds from white and black mustard grains: An optimization study for recovery and identification of phenolic antioxidants. *Industrial Crops and Products, 135*, 294–300.

Rastegar, S., Hassanzadeh Khankahdani, H., & Rahimzadeh, M. (2019). Effectiveness of alginate coating on antioxidant enzymes and biochemical changes during storage of mango fruit. *Journal of Food Biochemistry, 43*(11), Article e12990.

Ren, Y., Li, B., Jia, H., Yang, X., Sun, Y., Shou, J., Jiang, G., Shi, Y., & Chen, K. (2023). Comparative analysis of fruit firmness and genes associated with cell wall metabolisms in three cultivated strawberries during ripening and postharvest. *Food Quality and Safety, 7*, Article fyad020.

Rico, D., Martin-Diana, A. B., Barat, J., & Barry-Ryan, C. (2007). Extending and measuring the quality of fresh-cut fruit and vegetables: A review. *Trends in Food Science & Technology, 18*(7), 373–386.

Rodríguez, K., Ah-Hen, K., Vega-Gálvez, A., López, J., Quispe-Fuentes, I., Lemus-Mondaca, R., & Gálvez-Ranilla, L. (2014). Changes in bioactive compounds and antioxidant activity during convective drying of murta (Ugni molinae T.) berries. *International Journal of Food Science and Technology, 49*(4), 990–1000.

Ruelas-Chacon, X., Contreras-Esquivel, J., Montañez, J., Aguilera-Carbo, A., Reyes-Vega, M., Peralta-Rodriguez, R., & Sanchéz-Brambila, G. (2017). Guar gum as an edible coating for enhancing shelf-life and improving postharvest quality of roma tomato (Solanum lycopersicum L.). *Journal of Food Quality, 2017*(1), Article 8608304.

Santos, J. L. S., Faria, P. V., Gerson de Freitas, S. V., Gonçalves, G. A., & Machado, G. G. L. (2022). Application of Sodium Alginate edible coating incorporated with lemon grass (Cymbopogon citratus) extract in strawberry conservation. *Revista Iberoamericana de Tecnología Postcosecha, 23*(1), 27–45.

Sha, X., Si, X., Zhu, Y., Wang, S., & Zhao, Y. (2025). Automatic three-dimensional reconstruction of transparent objects with multiple optimization strategies under limited constraints. *Image and Vision Computing*, Article 105580.

Singleton, V. L., Orthofer, R., & Lamuela-Raventós, R. M. (1999). [14] analysis of total phenols and other oxidation substrates and antioxidants by means of folin-ciocalteu reagent. In *, 299*. *Methods in enzymology* (pp. 152–178). Elsevier.

Sotera, E. M., Rodriguez, A., Gamboa-Santos, J., & Campañone, L. A. (2021). Review characterization of edible films formulated with sodium alginate and low-methoxyl pectin in osmotic dehydration applications. *International Journal of Food Science and Technology, 56*(10), 4995–5006.

Syarifuddin, A., Muflih, M. H., Izzah, N., Fadillah, U., Ainani, A. F., & Dirpan, A. (2025). Pectin-based edible films and coatings: From extraction to application on food packaging towards circular economy-A review. *Carbohydrate Polymer Technologies and Applications*, Article 100680.

Trajkovska Petkoska, A., Daniloski, D., Kumar, N., Pratibha, & Broach, A. T. (2021a). Active edible packaging: A sustainable way to deliver functional bioactive compounds and nutraceuticals. In *Sustainable packaging* (pp. 225–264). Springer.

Trajkovska Petkoska, A., Daniloski, D., Kumar, N., Pratibha, & Broach, A. T. (2021b). Biobased materials as a sustainable potential for edible packaging. In *Sustainable packaging* (pp. 111–135). Springer.

Wang, C., Meng, L., Zhang, G., Yang, X., Pang, B., Cheng, J., He, B., & Sun, F. (2024). Unraveling crop enzymatic browning through integrated omics. *Frontiers in Plant Science, 15*, Article 1342639.

Wang, X., Su, H., & Liu, X. (2025). The impact of green technological innovation on industrial structural optimization under dual-carbon targets: The role of the moderating effect of carbon emission efficiency. *Sustainability, 17*(14), 6313.

Wani, S. M., Gull, A., Ahad, T., Malik, A., Ganaie, T. A., Masoodi, F. A., & Gani, A. (2021). Effect of gum Arabic, xanthan and carrageenan coatings containing antimicrobial agent on postharvest quality of strawberry: Assessing the physicochemical, enzyme activity and bioactive properties. *International Journal of Biological Macromolecules, 183*, 2100–2108.

Yazıcıoğlu, N., & Siyasal, K. (2024). Quick-dissolvable heat-sealable edible films made from orange peel powder and guar gum for instant beverage packaging. *Food and Health, 10*(2), 115–128.

Yu, S., Wang, Y., Chen, T., Li, M., Zhang, X., Huang, B., Xu, J., & Wang, G. (2025). An inclined groove and its optimization design method for improving the energy performance at the saddle zone of axial flow pumps. *Energy*, Article 136527.

Zhang, H., Pu, J., Tang, Y., Wang, M., Tian, K., Wang, Y., Luo, X., & Deng, Q. (2022). Changes in phenolic compounds and antioxidant activity during development of 'Qiangcuili'and 'Cuihongli'fruit. *Foods, 11*(20), 3198.